

10 POLYNOMIALS

Figure 10.1 The paths of rockets are calculated using polynomials. (credit: NASA, Public Domain)

Chapter Outline

10.1 Add and Subtract Polynomials
10.2 Use Multiplication Properties of Exponents
10.3 Multiply Polynomials
10.4 Divide Monomials
10.5 Integer Exponents and Scientific Notation
10.6 Introduction to Factoring Polynomials

Introduction to Polynomials

Expressions known as polynomials are used widely in algebra. Applications of these expressions are essential to many careers, including economists, engineers, and scientists. In this chapter, we will find out what polynomials are and how to manipulate them through basic mathematical operations.

10.1 Add and Subtract Polynomials

Learning Objectives

By the end of this section, you will be able to:

- Identify polynomials, monomials, binomials, and trinomials
- Determine the degree of polynomials
- Add and subtract monomials
- Add and subtract polynomials
- Evaluate a polynomial for a given value

BE PREPARED 10.1 Before you get started, take this readiness quiz.

Simplify: $8x + 3x$.
If you missed this problem, review Example 2.22.

BE PREPARED 10.2 Subtract: $(5n + 8) - (2n - 1)$.
If you missed this problem, review Example 7.29.

BE PREPARED 10.3 Evaluate: $4y^2$ when $y = 5$
If you missed this problem, review Example 2.18.

Identify Polynomials, Monomials, Binomials, and Trinomials

In Evaluate, Simplify, and Translate Expressions, you learned that a term is a constant or the product of a constant and one or more variables. The constant is called a coefficient. When it is of the form ax^m, where a is a constant and m is a whole number, it is called a monomial. A monomial, or a sum and/or difference of monomials, is called a polynomial.

Polynomials

polynomial—A monomial, or two or more monomials, combined by addition or subtraction

monomial—A polynomial with exactly one term

binomial— A polynomial with exactly two terms

trinomial—A polynomial with exactly three terms

Notice the roots:

- *poly-* means many
- *mono-* means one
- *bi-* means two
- *tri-* means three

Here are some examples of polynomials:

Polynomial	$b+1$	$4y^2-7y+2$	$5x^5-4x^4+x^3+8x^2-9x+1$
Monomial	5	$4b^2$	$-9x^3$
Binomial	$3a-7$	y^2-9	$17x^3+14x^2$
Trinomial	x^2-5x+6	$4y^2-7y+2$	$5a^4-3a^3+a$

Notice that every monomial, binomial, and trinomial is also a polynomial. They are special members of the family of polynomials and so they have special names. We use the words 'monomial', 'binomial', and 'trinomial' when referring to these special polynomials and just call all the rest 'polynomials'.

EXAMPLE 10.1

Determine whether each polynomial is a monomial, binomial, trinomial, or other polynomial:

ⓐ $8x^2-7x-9$ ⓑ $-5a^4$ ⓒ $x^4-7x^3-6x^2+5x+2$ ⓓ $11-4y^3$ ⓔ n

Solution

	Polynomial	Number of terms	Type
ⓐ	$8x^2-7x-9$	3	Trinomial
ⓑ	$-5a^4$	1	Monomial
ⓒ	$x^4-7x^3-6x^2+5x+2$	5	Polynomial
ⓓ	$11-4y^3$	2	Binomial
ⓔ	n	1	Monomial

TRY IT 10.1 Determine whether each polynomial is a monomial, binomial, trinomial, or other polynomial.

ⓐ z ⓑ $2x^3 - 4x^2 - x - 8$ ⓒ $6x^2 - 4x + 1$ ⓓ $9 - 4y^2$ ⓔ $3x^7$

TRY IT 10.2 Determine whether each polynomial is a monomial, binomial, trinomial, or other polynomial.

ⓐ $y^3 - 8$ ⓑ $9x^3 - 5x^2 - x$ ⓒ $x^4 - 3x^2 - 4x - 7$ ⓓ $-y^4$ ⓔ w

Determine the Degree of Polynomials

In this section, we will work with polynomials that have only one variable in each term. The degree of a polynomial and the degree of its terms are determined by the exponents of the variable.

A monomial that has no variable, just a constant, is a special case. The degree of a constant is 0—it has no variable.

Degree of a Polynomial

The **degree of a term** is the exponent of its variable.

The degree of a constant is 0.

The degree of a polynomial is the highest degree of all its terms.

Let's see how this works by looking at several polynomials. We'll take it step by step, starting with monomials, and then progressing to polynomials with more terms.

Remember: Any base written without an exponent has an implied exponent of 1.

Monomials	5	$4b^2$	$-9x^3$	-18
Degree	0	2	3	0

Binomial	$b + 1$	$3a - 7$	$y^2 - 9$	$17x^3 + 14x^2$
Degree of each term	1 0	1 0	2 0	3 2
Degree of polynomial	1	1	2	3

Trinomial	$x^2 - 5x + 6$	$4y^2 - 7y + 2$	$5a^4 - 3a^3 + a$	$x^4 + 2x^2 - 5$
Degree of each term	2 1 0	2 1 0	4 3 1	4 2 0
Degree of polynomial	2	2	4	4

Polynomial	$b + 1$	$4y^2 - 7y + 2$	$4x^4 + x^3 + 8x^2 - 9x + 1$
Degree of each term	1 0	2 1 0	4 3 2 1 0
Degree of polynomial	1	2	4

EXAMPLE 10.2

Find the degree of the following polynomials:

ⓐ $4x$ ⓑ $3x^3 - 5x + 7$ ⓒ -11 ⓓ $-6x^2 + 9x - 3$ ⓔ $8x + 2$

Solution

ⓐ	$4x$
The exponent of x is one. $x = x^1$	The degree is 1.
ⓑ	$3x^3 - 5x + 7$
The highest degree of all the terms is 3.	The degree is 3
ⓒ	-11
The degree of a constant is 0.	The degree is 0.
ⓓ	$-6x^2 + 9x - 3$
The highest degree of all the terms is 2.	The degree is 2.
ⓔ	$8x + 2$
The highest degree of all the terms is 1.	The degree is 1.

TRY IT 10.3 Find the degree of the following polynomials:

ⓐ $-6y$ ⓑ $4x - 1$ ⓒ $3x^4 + 4x^2 - 8$ ⓓ $2y^2 + 3y + 9$ ⓔ -18

TRY IT 10.4 Find the degree of the following polynomials:

ⓐ 47 ⓑ $2x^2 - 8x + 2$ ⓒ $x^4 - 16$ ⓓ $y^5 - 5y^3 + y$ ⓔ $9a^3$

Working with polynomials is easier when you list the terms in descending order of degrees. When a polynomial is written this way, it is said to be in **standard form**. Look back at the polynomials in Example 10.2. Notice that they are all written in standard form. Get in the habit of writing the term with the highest degree first.

Add and Subtract Monomials

In The Language of Algebra, you simplified expressions by combining like terms. Adding and subtracting monomials is the same as combining like terms. Like terms must have the same variable with the same exponent. Recall that when combining like terms only the coefficients are combined, never the exponents.

EXAMPLE 10.3

Add: $17x^2 + 6x^2$.

Solution

	$17x^2 + 6x^2$
Combine like terms.	$23x^2$

TRY IT 10.5 Add: $12x^2 + 5x^2$.

TRY IT 10.6 Add: $-11y^2 + 8y^2$.

EXAMPLE 10.4

Subtract: $11n - (-8n)$.

Solution

	$11n - (-8n)$
Combine like terms.	$19n$

TRY IT 10.7 Subtract: $9n - (-5n)$.

TRY IT 10.8 Subtract: $-7a^3 - \left(-5a^3\right)$.

EXAMPLE 10.5

Simplify: $a^2 + 4b^2 - 7a^2$.

Solution

	$a^2 + 4b^2 - 7a^2$
Combine like terms.	$-6a^2 + 4b^2$

Remember, $-6a^2$ and $4b^2$ are not like terms. The variables are not the same.

TRY IT 10.9 Add: $3x^2 + 3y^2 - 5x^2$.

TRY IT 10.10 Add: $2a^2 + b^2 - 4a^2$.

Add and Subtract Polynomials

Adding and subtracting polynomials can be thought of as just adding and subtracting like terms. Look for like terms—those with the same variables with the same exponent. The Commutative Property allows us to rearrange the terms to put like terms together. It may also be helpful to underline, circle, or box like terms.

EXAMPLE 10.6

Find the sum: $\left(4x^2 - 5x + 1\right) + \left(3x^2 - 8x - 9\right)$.

Solution

	$(4x^2 - 5x + 1) + (3x^2 - 8x - 9)$
Identify like terms.	$\underline{\underline{4x^2}} - \underline{5x} + \boxed{1} + \underline{\underline{3x^2}} - \underline{8x} - \boxed{9}$

Rearrange to get the like terms together.	$4x^2 + 3x^2 - 5x - 8x + \boxed{1} - \boxed{9}$
Combine like terms.	$7x^2 - 13x - 8$

TRY IT 10.11 Find the sum: $(3x^2 - 2x + 8) + (x^2 - 6x + 2)$.

TRY IT 10.12 Find the sum: $(7y^2 + 4y - 6) + (4y^2 + 5y + 1)$.

Parentheses are grouping symbols. When we add polynomials as we did in Example 10.6, we can rewrite the expression without parentheses and then combine like terms. But when we subtract polynomials, we must be very careful with the signs.

EXAMPLE 10.7

Find the difference: $(7u^2 - 5u + 3) - (4u^2 - 2)$.

Solution

	$(7u^2 - 5u + 3) - (4u^2 - 2)$
Distribute and identify like terms.	$7u^2 - 5u + \boxed{3} - 4u^2 + \boxed{2}$
Rearrange the terms.	$7u^2 - 4u^2 - 5u + 3 + \boxed{2}$
Combine like terms.	$3u^2 - 5u + 5$

TRY IT 10.13 Find the difference: $(6y^2 + 3y - 1) - (3y^2 - 4)$.

TRY IT 10.14 Find the difference: $(8u^2 - 7u - 2) - (5u^2 - 6u - 4)$.

EXAMPLE 10.8

Subtract: $(m^2 - 3m + 8)$ from $(9m^2 - 7m + 4)$.

Solution

	Subtract $(m^2 - 3m + 8)$ from $(9m^2 - 7m + 4)$
Distribute and identify like terms.	$9m^2 - 7m + \boxed{4} - m^2 + 3m - \boxed{8}$
Rearrange the terms.	$9m^2 - m^2 - 7m + 3m + \boxed{4} - \boxed{8}$
Combine like terms.	$8m^2 - 4m - 4$

TRY IT 10.15 Subtract: $(4n^2 - 7n - 3)$ from $(8n^2 + 5n - 3)$.

TRY IT 10.16 Subtract: $(a^2 - 4a - 9)$ from $(6a^2 + 4a - 1)$.

Evaluate a Polynomial for a Given Value

In The Language of Algebra we evaluated expressions. Since polynomials are expressions, we'll follow the same procedures to evaluate polynomials—substitute the given value for the variable into the polynomial, and then simplify.

EXAMPLE 10.9

Evaluate $3x^2 - 9x + 7$ when

ⓐ $x = 3$ ⓑ $x = -1$

Solution

ⓐ $x = 3$

	$3x^2 - 9x + 7$
Substitute 3 for x	$3(3)^2 - 9(3) + 7$
Simplify the expression with the exponent.	$3 \cdot 9 - 9(3) + 7$
Multiply.	$27 - 27 + 7$
Simplify.	7

ⓑ $x = -1$

	$3x^2 - 9x + 7$
Substitute −1 for x	$3(-1)^2 - 9(-1) + 7$
Simplify the expression with the exponent.	$3 \cdot 1 - 9(-1) + 7$
Multiply.	$3 + 9 + 7$
Simplify.	19

TRY IT 10.17 Evaluate: $2x^2 + 4x - 3$ when

ⓐ $x = 2$ ⓑ $x = -3$

TRY IT 10.18 Evaluate: $7y^2 - y - 2$ when

ⓐ $y = -4$ ⓑ $y = 0$

EXAMPLE 10.10

The polynomial $-16t^2 + 300$ gives the height of an object t seconds after it is dropped from a 300 foot tall bridge. Find

the height after $t = 3$ seconds.

Solution

	$-16t^2 + 300$
Substitute 3 for t	$-16(3)^2 + 300$
Simplify the expression with the exponent.	$-16 \cdot 9 + 300$
Multiply.	$-144 + 300$
Simplify.	156

TRY IT 10.19 The polynomial $-8t^2 + 24t + 4$ gives the height, in feet, of a ball t seconds after it is tossed into the air, from an initial height of 4 feet. Find the height after $t = 3$ seconds.

TRY IT 10.20 The polynomial $-8t^2 + 24t + 4$ gives the height, in feet, of a ball x seconds after it is tossed into the air, from an initial height of 4 feet. Find the height after $t = 2$ seconds.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Adding Polynomials (https://openstaxcollege.org/l/24addpolynomi)

Subtracting Polynomials (https://openstaxcollege.org/l/24subtractpoly)

SECTION 10.1 EXERCISES

Practice Makes Perfect

Identify Polynomials, Monomials, Binomials and Trinomials

In the following exercises, determine if each of the polynomials is a monomial, binomial, trinomial, or other polynomial.

1. $5x + 2$

2. $z^2 - 5z - 6$

3. $a^2 + 9a + 18$

4. $-12p^4$

5. $y^3 - 8y^2 + 2y - 16$

6. $10 - 9x$

7. $23y^2$

8. $m^4 + 4m^3 + 6m^2 + 4m + 1$

Determine the Degree of Polynomials

In the following exercises, determine the degree of each polynomial.

9. $8a^5 - 2a^3 + 1$

10. $5c^3 + 11c^2 - c - 8$

11. $3x - 12$

12. $4y + 17$

13. -13

14. -22

Add and Subtract Monomials

In the following exercises, add or subtract the monomials.

15. $6x^2 + 9x^2$

16. $4y^3 + 6y^3$

17. $-12u + 4u$

18. $-3m + 9m$

19. $5a + 7b$

20. $8y + 6z$

21. Add: $4a, -3b, -8a$

22. Add: $4x, 3y, -3x$

23. $18x - 2x$

24. $13a - 3a$

25. Subtract $5x^6$ from $-12x^6$

26. Subtract $2p^4$ from $-7p^4$

Add and Subtract Polynomials

In the following exercises, add or subtract the polynomials.

27. $(4y^2 + 10y + 3) + (8y^2 - 6y + 5)$

28. $(7x^2 - 9x + 2) + (6x^2 - 4x + 3)$

29. $(x^2 + 6x + 8) + (-4x^2 + 11x - 9)$

30. $(y^2 + 9y + 4) + (-2y^2 - 5y - 1)$

31. $(3a^2 + 7) + (a^2 - 7a - 18)$

32. $(p^2 - 5p - 11) + (3p^2 + 9)$

33. $(6m^2 - 9m - 3) - (2m^2 + m - 5)$

34. $(3n^2 - 4n + 1) - (4n^2 - n - 2)$

35. $(z^2 + 8z + 9) - (z^2 - 3z + 1)$

36. $(z^2 - 7z + 5) - (z^2 - 8z + 6)$

37. $(12s^2 - 15s) - (s - 9)$

38. $(10r^2 - 20r) - (r - 8)$

39. Find the sum of $\left(2p^3 - 8\right)$ and $\left(p^2 + 9p + 18\right)$

40. Find the sum of $\left(q^2 + 4q + 13\right)$ and $\left(7q^3 - 3\right)$

41. Subtract $\left(7x^2 - 4x + 2\right)$ from $\left(8x^2 - x + 6\right)$

42. Subtract $\left(5x^2 - x + 12\right)$ from $\left(9x^2 - 6x - 20\right)$

43. Find the difference of $\left(w^2 + w - 42\right)$ and $\left(w^2 - 10w + 24\right)$

44. Find the difference of $\left(z^2 - 3z - 18\right)$ and $\left(z^2 + 5z - 20\right)$

Evaluate a Polynomial for a Given Value

In the following exercises, evaluate each polynomial for the given value.

45. Evaluate $8y^2 - 3y + 2$
ⓐ $y = 5$ ⓑ $y = -2$
ⓒ $y = 0$

46. Evaluate $5y^2 - y - 7$ when:
ⓐ $y = -4$ ⓑ $y = 1$
ⓒ $y = 0$

47. Evaluate $4 - 36x$ when:
ⓐ $x = 3$ ⓑ $x = 0$
ⓒ $x = -1$

48. Evaluate $16 - 36x^2$ when:
ⓐ $x = -1$ ⓑ $x = 0$
ⓒ $x = 2$

49. A window washer drops a squeegee from a platform 275 feet high. The polynomial $-16t^2 + 275$ gives the height of the squeegee t seconds after it was dropped. Find the height after $t = 4$ seconds.

50. A manufacturer of microwave ovens has found that the revenue received from selling microwaves at a cost of p dollars each is given by the polynomial $-5p^2 + 350p$. Find the revenue received when $p = 50$ dollars.

Everyday Math

51. **Fuel Efficiency** The fuel efficiency (in miles per gallon) of a bus going at a speed of x miles per hour is given by the polynomial $-\frac{1}{160}x^2 + \frac{1}{2}x$. Find the fuel efficiency when $x = 40$ mph.

52. **Stopping Distance** The number of feet it takes for a car traveling at x miles per hour to stop on dry, level concrete is given by the polynomial $0.06x^2 + 1.1x$. Find the stopping distance when $x = 60$ mph.

Writing Exercises

53. Using your own words, explain the difference between a monomial, a binomial, and a trinomial.

54. Eloise thinks the sum $5x^2 + 3x^4$ is $8x^6$. What is wrong with her reasoning?